

Week 4 Day 3: Summary

Feature Engineering, Cross-Validation & Model Comparison

Introduction

The goal of this task was to improve the Adult Income prediction model by creating new features, comparing multiple machine learning models using 5-fold cross-validation, and identifying the best model for future hyperparameter tuning. Precision remained the primary evaluation metric because the focus is on correctly identifying high-income individuals while minimizing false positives.

1. Engineered Features

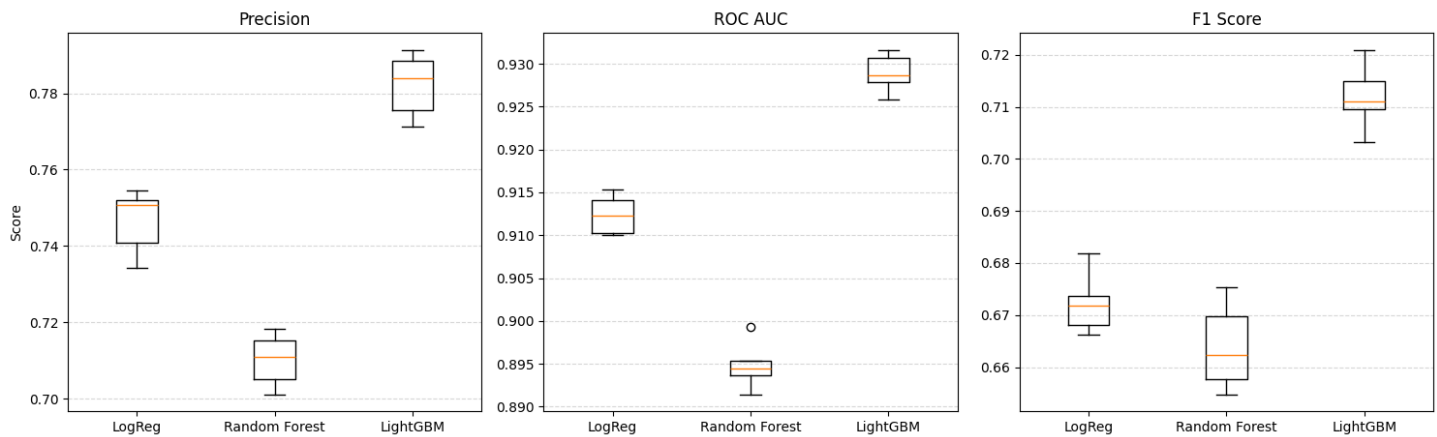
Six new features were created using only the information available in each row, avoiding any data leakage.

	Feature	Type	Creation Rule	Why it may help
0	age_bucket	Categorical	Age grouped into ranges	Captures career stages
1	hours_bucket	Categorical	Hours grouped into ranges	Captures work intensity
2	capital_gain_flag	Binary	capital-gain > 0	Identifies investment income
3	log_capital_gain	Numeric	log1p(capital-gain)	Reduces skew
4	higher_education	Binary	education-num >= 13	Identifies highly educated people
5	edu_hours	Numeric	education-num × hours-per-week	Captures interaction between education and wor...

2. Cross-Validation (CV) Model Comparison

Model	Precision	ROC AUC	F1
Logistic Regression	0.747 ± 0.008	0.912 ± 0.002	0.672 ± 0.005
Random Forest	0.710 ± 0.006	0.895 ± 0.003	0.664 ± 0.008
LightGBM	0.782 ± 0.008	0.929 ± 0.002	0.712 ± 0.006

5-Fold Cross-Validation Scores



LightGBM performed the best across all evaluation metrics. It achieved the highest Precision, ROC AUC, and F1 score while remaining consistent across all five folds. Logistic Regression also performed well but was slightly behind LightGBM, whereas Random Forest had the weakest overall performance.

3. Statistical Test

A statistical comparison was performed between the two best models (LightGBM and Logistic Regression) using a paired test on Precision scores across the five folds.

```
t-statistic: 17.762
p-value: 0.0001
```

The p-value is 0.0001 making it below 0.05, so the difference between LightGBM and Logistic Regression is statistically significant, so we found a better model that is backed-up! This means LightGBM's higher Precision is unlikely to be due to chance, making it the better model.

4. Feature Importance

LightGBM showed `edu_hours` as the most important engineered feature. Logistic Regression gave the largest positive coefficient to `log_capital_gain`. Age buckets and hours buckets also helped, which matches the relationship between income, education, work hours and investment income.

5. Feature Selection

A SelectKBest feature selection step using Mutual Information was tested before training LightGBM.

	Model	Precision	ROC AUC	F1	Training Time (s)
0	Original LightGBM	0.782150	0.928927	0.711923	-
1	Feature Selection	0.781809	0.925838	0.694380	79.97

Feature selection kept most of the engineered features, including `edu_hours`, `log_capital_gain`, `higher_education`, `capital_gain_flag`, `age_bucket`, and `hours_bucket`. However, it slightly reduced model performance while adding extra processing time. Because of this, using all available features is the better choice for the next stage.

Recommendation for Hyperparameter Tuning

I will continue with **LightGBM** for hyperparameter tuning because it achieved the best overall performance (Precision 0.782, ROC AUC 0.929, F1 0.712). Keep the **full feature set** because SelectKBest slightly reduced performance. In particular, I'll retain `edu_hours`, `log_capital_gain`, `higher_education`, `capital_gain_flag`, `age_bucket` and `hours_bucket`.